

Reflection Essay

CSO-610 Summer 2025-26 — Task 02

What I was trying to do

The goal was to estimate real-world distances from a single photograph — without any depth sensor, stereo camera, or known camera intrinsics. The technique, called *single view metrology*, feels almost like cheating: you take one flat image and somehow extract 3D information. But there is a catch, and the catch is the whole point of the assignment. You can only do it if you already know some distances. The math does not create information; it transfers it from known points to unknown ones.

The concept that clicked

Cross-ratio was the hardest concept to internalise. In class we defined it as a ratio of ratios of signed distances between four collinear points. What made it feel abstract was the word “preserved” — preserved under what, exactly?

The penny dropped when I thought about it as follows. A camera projection is a perspective map from the 3D world to a 2D image plane. This map destroys almost everything: lengths, angles, area, parallelism. But it preserves cross-ratio. So if I measure the cross-ratio of four points in the image, that number is *identical* to the cross-ratio of the corresponding four real-world points. That is my bridge from pixels to metres.

The vanishing point simplification also clicked only after working through the limit. When the fourth point V goes to infinity, the cross-ratio

$$\text{CR}(A, B; C, V) = \frac{(A - C)(B - V)}{(B - C)(A - V)} \rightarrow \frac{A - C}{B - C} = \frac{AC}{BC}$$

because the V terms cancel. This gives one equation, one unknown. It is elegant but only works if you can correctly identify the vanishing point in the image — which turns out to be harder than it looks.

The practical parts that were hard

Running YOLOv8 was not hard once I found the right documentation. The hard part was getting it to *work*. With the default 640-pixel input, the model found nothing on a 4032-pixel-tall image where the bicycle occupies about 60 pixels. I wasted time checking class IDs before discovering the `imgsz` argument.

The pixel annotation was meditative but error-prone. MS Paint shows coordinates in the status bar, which is useful, but your mouse has to be exactly on the right feature. The vanishing point is particularly tricky because it lies outside the area of the image where the road is visible — it is where the road edges would meet if extended, and that meeting point is near the sky. I estimated it from the arch position and the general convergence of the footpath edges, but I know my estimate has a few hundred pixels of uncertainty.

Taking the personal photos

I took photos in a narrow residential lane near my home. This was actually a good choice for the geometry: the walls on both sides are close enough that their convergence to the vanishing point is clearly visible within the frame. On a wide road, the vanishing point is much further away and the convergence is less obvious.

The main challenge was that the lane had parked scooters on both sides, which made finding an unobstructed view of my subject difficult. I took six photos before getting one where YOLO detected the person confidently (confidence 0.68). The first attempts had the person partially behind a scooter.

I also tried taking a photo at dusk. Detection confidence dropped to around 0.45. The bounding box bottom was less reliable because the shadow extended beyond the person's feet. This was a useful reminder that YOLO is not lighting-invariant even though the cross-ratio method itself is.

What I would do differently

If I were to redo this assignment, I would place clear physical markers (tape or chalk) at measured distances before taking the photo, rather than trying to identify reference points from existing features (like a manhole cover). The manhole cover was convenient but I had to estimate its distance by stepping it out, which is less accurate than a measuring tape.

I would also use a small tripod and take the photo from ground level, which gives a more natural perspective with the vanishing point visible at a comfortable height in the frame. Handheld photos at eye level put the vanishing point near the centre-top of the image, which is fine, but any slight camera tilt changes the vanishing point significantly.

Summary

This assignment connected three ideas I had treated separately before: projective geometry (from the lectures on homogeneous coordinates), deep learning object detection (YOLO), and physical measurement. Using them together felt more like engineering than pure computer vision. The cross-ratio is a beautiful theoretical tool, but making it work in practice requires care at every step: the right input resolution for the detector, the right bounding-box reference point, and careful annotation of reference pixels. That gap between “mathematically elegant” and “practically correct” is, I think, the real lesson of this task.